

WASP-10b

R-0920

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_170622 · created 2026-07-29T08:44:13+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457926.8674 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.06
Transit depth (Rp/Rs)^2	3.62 +/- 2.3 [%]
Orbital Inclination (inc)	84.64 +/- 1.87
Airmass coefficient 1 (a1)	1.1 +/- 0.32
Airmass coefficient 2 (a2)	-0.007 +/- 0.033
Scatter in the residuals of the lightcurve fit is	30.47 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.041 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.06 vs 0.1592 (deviation 0.51σ)
Duration fit vs published	0.041 d vs 0.0946 d (deviation 1.53σ)
Mid-time O–C	1.7 min
Depth SNR	3.2
Residual scatter	30.47 %

Light curve

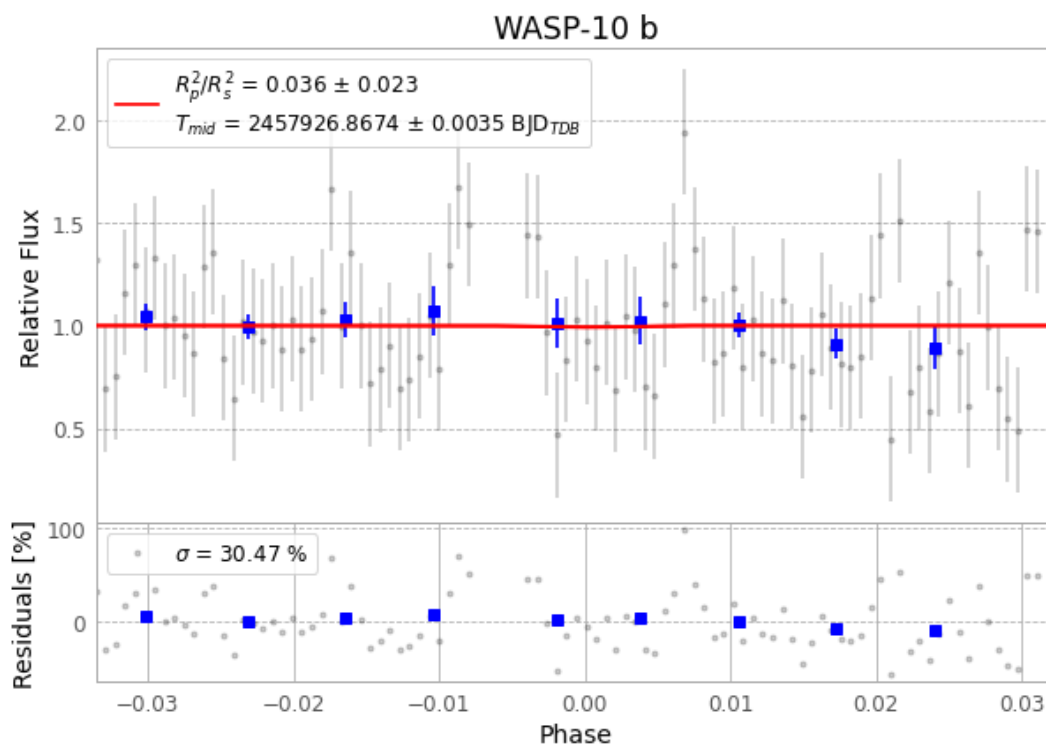


Figure 1 — FinalLightCurve_WASP-10 b_2017-06-21.png (SHA-256 2c2d0bf9b9fd40b2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 434], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3b2475fdb0ca914...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

98 FITS frames (64 MB), WASP-10170622061814.FITS → WASP-10170622111212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-170622.log (df063e110387...), FinalLightCurve_WASP-10 b_2017-06-21.png (2c2d0bf9b9fd...), FinalLightCurve_WASP-10 b_2017-06-21.csv (a55504b6174d...)

Compute resources

Wall time 210.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 08:44Z.